

ML Model: Rented Bikes Demand in Seoul

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Abstract—The procedure of this project starts with an ETL method for raw data treatment, followed by an EDA for a better understanding of the situation and of the limitations imposed by the data split. Model development begins with a linear regression implemented from scratch, using batch gradient descent, which serves as a transparent baseline and establishes what a purely linear relationship can achieve on this problem. The exploratory analysis reveals cyclical and non-linear behaviours that a linear model cannot represent; a second model is therefore built using a framework, and both are compared on the same subsets and metrics to determine which better fits the nature of this demand-prediction problem., then comparing the results.

Index Terms—Machine Learning, ETL, EDA, PCA, Seoul Bike Sharing Demand, Prediction Model

I. INTRODUCTION

It's possible to make a prediction by pure statistics; however, the first results on this project look forward to taking advantage of machine learning and producing an accurate prediction. Although the development and results for this project are limited and confined by the sandbox of raw data, it should be a first step in understanding an introduction to machine learning for me and the reader, an exercise of the mind so better knowledge can be produced. And if someday it were to create an absolute model that could predict the demand with such precision for bikes, transportation in Seoul or other cities, this and all people working on the same data and similar projects would be the foundation and first step for it.

II. EXTRACT, TRANSFORM, LOAD (ETL)

A. Data Extraction, Initial Inspection and Understanding

To produce an ETL we need to understand the raw dataset. The dataset comprises 14 features and 8760 instances in particular. We are talking about 365 days from 1 December 2017 to 30 November 2018, divided by hour instances; besides, out of those 14 features, the goal of the project is to predict the feature Rented Bike Count given the other parameters.

Furthermore, for these variables is important to understand that: In the feature Visibility, the method of measurement or instrument is unknown, but every unit is expressed as "10m" referring to a longitudinal distance in which it is visible, probably due to weather factors. The Dew point is the temperature at which air becomes fully saturated with water vapor and cannot hold moisture any longer. Is important when predicting weather forecasts due to its relation with other factors. Solar Radiation is measured in MegaJoules per square meter and represents the total accumulated energy produced by the sun in a lapse of time (Hour in this case). The Date its

TABLE I
RAW DATASET DESCRIPTION

Feature	Description	Type
Date	Format dd/mm/yyyy	Object
Rented Bike Count	Total bikes rented per hour (Target variable)	Integer
Hour	Hour of the day (0–23)	Integer
Temperature	Ambient temperature (°C)	Float
Humidity	Relative humidity (%)	Integer
Wind speed	Wind speed (m/s)	Float
Visibility	Visibility distance in 10m units	Integer
Dew point temperature	Temperature at which air becomes fully saturated (°C)	Float
Solar Radiation	Accumulated solar energy (MJ/m^2)	Float
Rainfall	Rainfall amount (mm)	Float
Snowfall	Snowfall amount (cm)	Float
Seasons	Seasonal periods of the year	Object
Holiday	Holiday indicator (Holiday / No Holiday)	Object
Functioning Day	Operational status of the rental system	Object

resourceful to understand results and the chronology of events; however, it won't be fed to the model since it isn't helpful for a prediction, just for a search. Worth mentioning that it had no null values for any feature.

B. Understandable Data for the Model

The first step is the transformation of all Objects and Strings into understandable data. For Holiday, we transformed it into a binary integer value; it either is a holiday or isn't. Seasons were transformed into an integer variable, 1-4 starting with Winter to try and make an numerical distinction. Functioning day was also transformed into a binary integer value

C. Visibility Uncertainty Translation

For the visibility uncertainty on how it was measured, it was translated into a ratio of how much was visible for users when riding a bike, where 0 is no visibility and 1 is the maximum visibility; this was for better digestion for our model. This could be done because, after data inspection, it had a maximum value of 2000 and referred to a distance value, so it could be scaled.

D. Functionality Imputation

By nature, although having no null values, non-functioning days had a rented bike count of 0. After inspection, there were 295 non-functioning hours that represent approximately

3.36% of the dataset, and only these 295 instances presented an absolute 0 rented bike count. Due to the assumption that in the dataset there is not another possible combination that gives a rented bike count of 0 and represents a small amount of information, it was decided to perform an imputation of the rented bike count for these instances. The method for imputation was interpolation of the data. The interpolation was calculated as the rounded mean value for the same hour from the closest functioning instance before and after. This imputation takes advantage of the limited data; it cannot be used to train because we don't want the model to find a 100% correlation pattern on an obvious statement non-functioning days=0rented bikes and therefore influence the rest of the values, but it also can not be used in testing, as it doesn't have anything to compare to.

E. Date Feature Compensation

As the date won't be fed to compensate the feature, we can extract a binary value that can help our model better understand the routines. From the date, using pandas pd.to_datetime, we can get an int binary value for the weekend feature.

F. ETL Dataset

Up to this point, we do not possess null instances across all the data. The data has been transformed into digestible, understandable units and values. All calculations for the new data were successful and do not contain errors. The data was treated to the optimal type. Finally, all instances are still in chronological order. We end with this post-ETL dataset.

TABLE II
POST ETL DATASET

Feature	Type	Feature Class
Date	Value	Map
Rented Bike Count	Integer	Y
Hour	Integer	X
Temperature	Float	X
Humidity	Integer	X
Wind speed	Float	X
Visibility	Float,Ratio	X
Dew point temperature	Float	X
Solar Radiation	Float	X
Rainfall	Float	X
Snowfall	Float	X
Seasons	Int	X
Holiday	Int,Binary	X
Functioning Day	Int,Binary	MAP
Weekend	Int,Binary	X

G. Division of classes

We can now produce our division of classes for the model. The division percentage is set to be 90% of the data destined to train and the remaining 10% to test and validation of the model. To split the data, we have to consider the factor known as seasonality or trend. In finance and economics is

best noticed by the annual seasons in which data patterns emerge and behave differently between seasons. Therefore, it was decided that the division of data will be divided equally from all seasons. 36 days, representing approximately 10% of the total data, will be taken from the last days of each season, meaning that for each season the last 9 days, or 216 instances, will be destined for testing. The Y train and test subset will contain only Rented Bike Count; the X subset will have the rest, excluding the Date and Functioning Day. Recycling these features of data we create also a subset defined as Map that will contain these reused excluded features, as well as all the instances in order with the corresponding season, the subset it belongs to (train or test), and a local ID for the subset. This is with the purpose of having a map that will act as a tool when processing results, to graph all data in chronological order and explain possible anomalies due to the imputation or division of continuous data. The justification for creating the map is that it provides the information to order and filter data. After the division, we are left with the next subsets of data. Finally the remaining 10% will be split into 120 instances for test and 96 for validation, specifically test gets the last 5 days of the season and validation the 4 days before this.

TABLE III
REPRESENTATION OF DIVISION OF DATA

Class/Subset	Description
X test	Last X features from the last 120 instances from each season
Y test	Last Y features from the last 120 instances from each season
X val	First 96 X features from the last 216 instances from each season
Y val	First 96 Y features from the last 216 instances from each season
X train	Left X features from the split from each season
Y train	Left Y features from the split from each season
Map	All map features sorted by an index, keeping chronological order

III. EXPLORATORY DATA ANALYSIS (EDA)

The EDA of interest only falls under the training features, and in this occasion we will include our dependent variable. At this point, we can produce copies of our subset to model at our will without losing the information for the model.

A. Distribution with Histogram

For the analysis of histograms, we discriminate the data for Hours, Seasons, and Weekend, as that categorical data frequency won't add information. We then can produce of all histograms for all continuous data, including the histogram for Holiday. We are left with the following graphs.

1.1 The distribution for the Rented Bike Count is shown as a positively skewed distribution with a long tail. Most of the data falls below 500; we can assume that most instances or hours present fewer than 500 rentals.

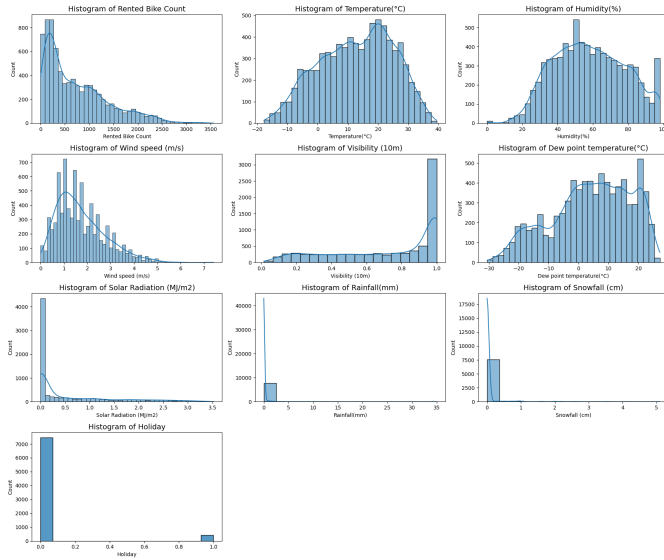


Fig. 1. Histograms of continuous variables and holiday distribution.

1.2 The distribution of Temperature seems a bit symmetrical, with a higher frequency at approximately 20 degrees. Most instances during the year of data collected were about 20 degrees Celsius in Seoul.

1.3 The distribution of humidity, although it almost looks symmetrical it has a high bias towards high values, with only a few occurrences around 0% and a high peak at 100%. With peaks at 50 and 100, we can assume that most instances presented at least a slightly higher humidity percentage in the data collected in Seoul.

1.4 The wind speed distribution presents a positively skewed distribution with a long tail distribution with most values falling around 1m/s. The majority of instances presented less than 3m/s wind speed readings in Seoul.

1.5 The distribution of visibility is highly negatively skewed, with the most frequent value being 1 also understood that the majority of instances present 100% or complete of visibility.

1.6 The distribution of Dew temperature seems to present a non-symmetrical distribution, with no high bias for the left side. It presents two pinnacles: first at around -20 degrees, then falls into a valley around -10 and rises to a higher pinnacle and larger group of frequency around 10 degrees, with a noticeable single peak at 20 degrees. We can assume that it presents a bimodal characteristic and that two groups of data could form around -20 degrees and 10 degrees of dew point; the dew point is ultimately noticed by people as it has a highly physical relation with weather. Therefore, there could be the possibility that it could provide a high numerical relationship when comparing the output with 2 possible classes. It could also be useful that the 20 degree peak has a strong numerical relationship with Temperature.

1.7 The histogram of solar radiation presents a positive skew frequency, with most of the data falling near 0 Mj/m²; most instances in Seoul present around 0 units of solar radiation, also interpreted as most of the time there are no

high readings of energy. This could be a highly noticeable remark to drop the information, as most information containing 0 as a value could not contribute to the model. However, the 0 reading could be explained by the recorded instances happening at night time or being a characteristic of Seoul. We can also expect the possibility of the model creating a relationship between this and other features, meaning that this data could identify possible groups in this highly skewed data and therefore is kept and will be fed to the model.

1.8 The histogram on rainfall shows most of the values fall near 0, below approximately 2.5. The distribution could be considered to not be a feature to study or even transformed as most of data falls on a small range however we have outliers and a maximum record of 35.0 millimeters in rain, these outliers of high rain instances could provide still a high chance influence in predicting, meaning with small data it can be almost absolute assumption that high rainfall produces very low rental behaviour therefore is kept for the model.

1.9 The histogram on the snowfall follows a similar behaviour, with most of the data falling below 0.5 cm and, due to the same logic, it could be argued that this information can be dropped or transformed; however, the maximum value occurring at 5.1 cm also gives the possibility of a high indicator of rental behaviour. These outliers could play an important role in the prediction model; therefore, it's kept.

1.10 The histogram on holidays was produced to understand a visual representation of the ratio between non-holiday or workless days. These days are important as it highly impacts the routine of the citizens of Seoul. Important to mention that this data can be fed without transformation; this is a binary value, so there is no use in exploring deeper into the distribution and mean; therefore kept as is for the model. I do propose the consideration that it could be transformed or grouped with more information, or even dropped, to explain, in a deeper manner, the behaviour of holidays. Meaning that holidays could impact in a highly unpredictable way as is, since holidays could produce high rental behaviour or even a lower one due to the possibility on lower citizens being in Seoul during these instances.

B. Distribution with Boxplot

We now proceed to create box-and-whisker graphs of box and whiskers for all continuous variables; graphing for categorical and discrete variables is not worth doing. We are left with the next graphs. It will be useful to visualize possible outliers better that could not be well noticed on the histogram graphs.

2.1 To describe it, the box itself is located on the left side, and the median is around 500 counts on the left side of the box. Regarding distribution, it has wide whiskers or spread, with high outliers. The worth-mentioning information to spot on the Boxplot for the Rented Bike Count is the high number of outliers. This could be an early remark that many real possible specific combinations of X features can provide high rental behaviour. of outliers.

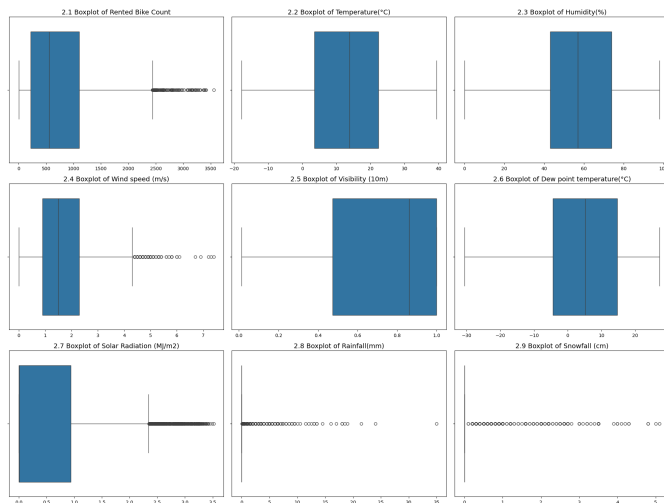


Fig. 2. Boxplot of continuous variables and holiday distribution.

2.2 , 2.3 , 2.5 , 2.6 After inspecting the boxplots produced for Temperature, Humidity percentage, Visibility and Dew point temperature, we can state that there are no noticeable outliers. Describing the graphs, it shows a highly packed box distribution in the data. The thing to consider is that the median for Temperature its notices it falls around 15 degrees and is located on the half of the box, making it a symmetrical distribution. For Humidity, the median falls around 55%, and it is more on the left side of the box. For Visibility, the length of the box is bigger but still skewed back, with the median located at around 0.85 on the right side of the box. Finally, for Dew point, the box is heavily packed and the median cuts symmetrically through the box, falling around 8 degrees. Even with large whiskers, the lack of outliers and a highly packed box is a sign of a more balanced or spread-out distribution. It indicates that the data is well scattered and most of the data has fallen within a statistical scope. It is important to take this factor in consideration when analysing further data also open to the possibility of falling by nature into a normal distribution, at this point the histograms 1.2, 1.3, 1.5, 1.6 didnt presented a strong characteristic of normality however it could be an early sign that this is due for the limitations on the data sample and that having a larger sample size of more than 1 year it could be appreciated a normal distribution then having solid new information on how to treat and transform this variables.

2.4 , 2.7 , 2.8 , 2.9 To describe Wind Speed, the box is located on the left side with a median around 1.5; the box almost looks symmetrical, but the median falls more to the left side. The Solar radiation Box has the median close to the left end of the box, indicating a high concentration of data around 0.0. Is worth mentioning that, in the case of Rainfall and Snowfall, the Box and whiskers are present in this graph; this is because it is a highly concentrated and skewed distribution, so both boxes and whiskers are small and located on the left side of their corresponding graph. Important to keep in mind when analysing further data. When checking the whiskers for

Wind Speed. Solar Radiation, Rainfall, and Snowfall present a high number of outliers. As mentioned before, all of these points do not mean they are negative characteristics for the model, and they will not spoil it; they are real possible events that could happen in Seoul. The purpose is not to eliminate or impute new values to these outliers, but to understand how they could affect the results of the model and how to manage and interpret the data, as the mean and variance for these values will have another perspective.

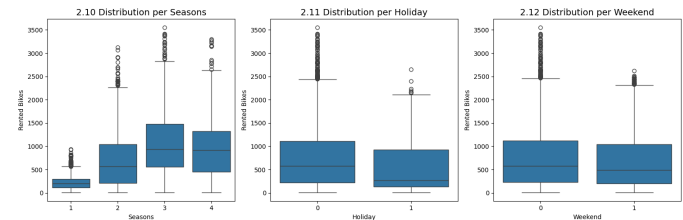


Fig. 3. Boxplot of categorical features.

2.10 Show us that each season contains many outliers for each season, and theoretically it's not ideal, but we can break it down. For winter, these outliers can be acceptable and understandable; the instances are by hour, and the difference in factors between early hours and midday can be drastically then it doesn't tell us that it will be extreme to predict. Besides, the boxplot itself seems very compact, and the whiskers aren't large sustaining that the dispersion still can be acceptable. For Autumn it contains it seems it contains a few outliers out of them all, and this can also be acceptable; the whiskers, even if they are large, can still be acceptable. The box itself is large, and the mean splits the center of the box; once again, it can be nice dispersion. For Spring and Summer, it does rises the suspect if the rental foo bikes spike or are widely distributed; this is because its appreciated the high number of outliers and the long whiskers. The box also falls far down from the highest values; that could be a sign that the wide dispersion could make our linear regression model difficult.

2.11 Give us a great result even if the non-holidays seem to be more dispersed due to the highly number of outliers; the holiday category does seem to be more docile to our understanding and could be a sign that the model could find a pattern recognizing better on holidays. We shall also recognize that the holiday itself, even if it seems chaotic, compresses each instance of hours during a day and many more data; therefore, it's not a full sign it will be that chaotic for the model.

2.12 Take into consideration that the variable weekend was created; this boxplot gives the first insight if the variable itself works. It does tell the same story as the distribution per holiday; this could be evidence that the variable categorizes important behaviours; however, we should dig deeper to prove this.

C. Behavior through line plots

We will average every hour's demand across all of the train subset for every hour in the day. The purpose of this graph is

to know if this feature can be useful. Reading the graph, we can notice some similarities across early hours and the middle of the day; however, the spike readings on weekend days with high demand at 8 am and 18 pm hours create the importance of maintaining this feature. Therefore, this feature won't be dropped.

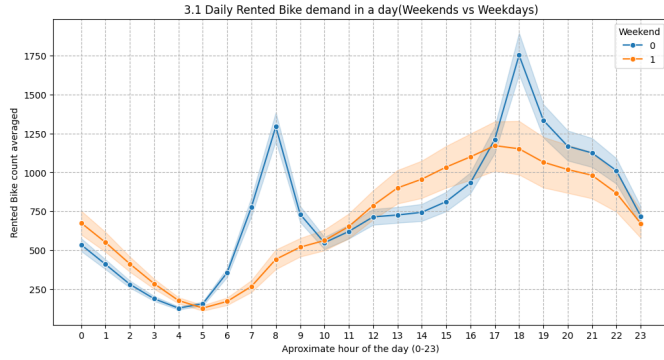


Fig. 4. Average demand for bikes during a day regarding the weekend feature.

We can apply this code logic and line plots for every feature and tell, with the best perception, how the behaviour is affecting the rental bikes across an average day for each feature. This is an important visualization that lets us better comprehend the individual effect it has. The nature of the code needs a metric and to declare classes and bins to distinguish continuous data. Therefore, I adjusted an approximation so each value of each feature can belong to a corresponding class. This is trying to imitate the fuzzy class concept, but the implementation criteria were different for each. The raining, solar radiation and snow turn them into binary outputs to create a more easier visual graph production and a more general depiction of the behaviour, if any of this category has a 0 value then is not happening this category and the opposite of that so we can illustrate the bike demand for when one of this feature is either not happening or happening by hour in an average day. For the categorical data, such as Holiday and Seasons already have values that can be grouped into classes for the line plot, and the other continuous data was matched into 3 classes that describe a minimum reading for the feature, a mild reading, and a high value for each of these features. The bins are established by the reading of the maximum and minimum values.

General findings In all days, no matter what the feature is or the values of the features, it always appears the same two spikes about 8-9 and 17-18, except for the holiday graph. This surely marks the behaviour of the labour schedule that heavily impacts the prediction of the bikes. To not leave things pending, I double-checked, and by different means I corroborated that the usual work shift in Seoul is a 9 to 18 hour shift. With this finding, I do raise the consideration that it would be highly impactful to introduce a way it has more importance to these instances, choose a model that can perceive and distinguish this hourly peak and not generalize across instances or create another split arrangement.

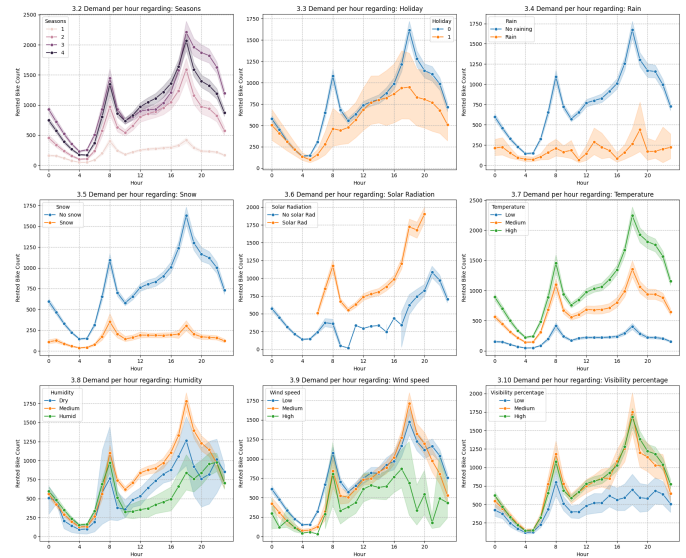


Fig. 5. Average demand for bikes during a day regarding other features.

3.2 , 3.3 The behaviour in the seasons and seasonality effect already gave us information; however, this graph depicts a different story. Zooming in on the average for every hour in the day tells us that, with differences, winter has less demand overall. On an average day, spring, summer, and autumn look similar, with a few hundred bikes' difference. The holiday days does confirm there is an effect for rentals: the non-working days have less demand and soften the hourly workshift peaks.

3.4 , 3.5 , 3.6 For rain, snow, and solar radiation, it gives us an approach to the expected behaviour. On average, rainy instances pick up fewer rented bike counts compared to non-rainy instances, and this is the same behaviour for snowing vs not snowing instances. Now, to explain the radiation, it now shifts the no solar radiation instances have, on average, fewer rental bike counts, and radiation days do have more. What is key to understand is that this is binary; it doesn't tell us how much radiation is fluctuating specifically, so it is plausible that on non-solar radiation instances there is less count; it's because it can be correlated with heavy, windy, cold, or rainy bad weather. What we do can learn is that, from solar radiation instances, it could possibly match better if we understand better the average daily cycle of the solar radiation; comparing the A.1 graph, we can visualize that after the greatest peak of solar radiation happening at approximately 13 hours, it starts to grow the demand on bikes as it falls down.

3.7 , 3.8 , 3.9 , 3.10 From temperature, humidity, wind speed, and visibility percentage, we observe three classes behaving similarly, with some important phenomena interesting to explain. I realized that, similar to solar radiation, it is also expected that Temperature can behave in a similar cyclic behaviour, so I created an average daily temperature cycle graph to realize that it could be explained and does fit the story that the high temperatures around 3 pm and the downfall to a low, not extreme temperature at noon fit the growing tendency of the rental demand in the same time lapse. For humidity,

the most notable characteristic is that, in the early hours, the difference in demand between classes is similar, but after 9 am, medium humidity has the majority of demand, followed by the driest instances and then the expected humid instances. Then, for wind speed, all instances in bike demand are close to each other, but it seems that after 4 pm, high wind speed gives lower levels of bike count. Then, for visibility, it seems that low visibility does give, in general, lower demand behaviour after 8 am, even decreasing at the end of the work shift to a difference of approximately a drop of 1000 counts.

D. Time series analysis

The general analysis on the time series for the rental bike was at first just a proposal to prove this dataset did have seasonality; however, it was important to re-graph it after the split to better understand what we were taking away from the train subset, as the important thing is to keep the most representative data. The comparison was made through line plot and scatter plots.

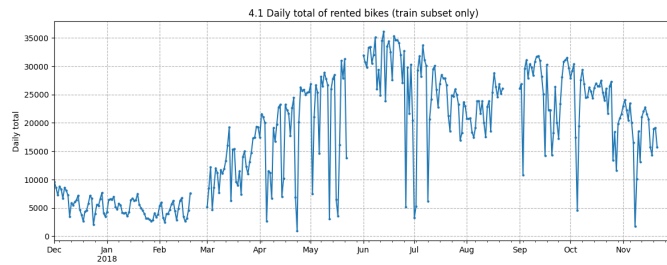


Fig. 6. Daily rental count across a year for train subset.

4.1 We will now analyse the average daily bike demand over the year of data for only the Y train subset, leaving the gaps for test and val. This is made simpler through the map subset to know what periods of time in order to graph. It is now possible to appreciate a difference between seasons in the daily demand. Winter (December through March) has the lowest daily average, and Summer (June through September) the highest average daily peaks of the year. Describing the whole year: December starts slow, and March starts growing with considerable large downfalls, but the growth gets to the max peak in June; then it starts to descend to August around 1000 rentals and then rise through September and October before falling again, arriving at November.

Then we arrived at the comparison with the scatter plot and all subsets, and I noticed some outliers from the non-training subsets, which created this next section.

E. The elephant in the room. Outliers in the split problem.

The question I proposed was ¿Does the split of the subsets take away important information to train the model?

To understand this, we can take the 4.2 graph and visualize clearly the contents of each subset. In comparison, in the 4.1 graph, we disregard the test and validation subsets so we can better appreciate that there are some big outliers that break a pattern, with huge sudden changes increasing or decreasing

considerably from one day to the next. Meaning that any other split considered could also have the same problem of low accuracy due to the model lacking these outliers to train and having problems predicting or training with these outliers and changing the parameters' final adjustments. We then produce

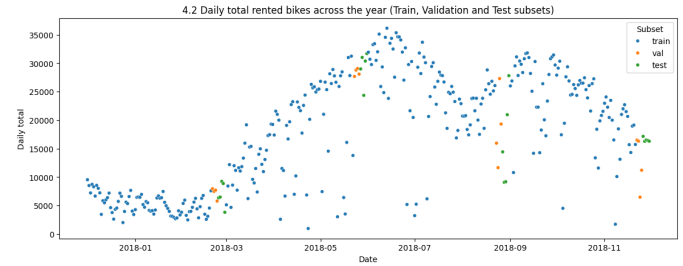


Fig. 7. Daily rental count across a year for all subsets.

the 4.2 graph comparing all three subsets, we can notice most of the outliers that the train doesn't get will happen around September. This can be acceptable due to the majority of data that won't predict do adjust better to the noticeable pattern. We have to take into consideration that this is only for the resulting data, or the data we want to predict, so we only used it as a first indication of possible bias by omission based purely on the results. In order to fully conclude whether there can be this danger, we have to study the X features further and notice that there is actual information left behind. So far, we can notice that after September there will be a high amount of anomalies; we then ask the question ¿Is there any of the data that justifies those changes? If so, ¿can we prove it can be picked up in the same pattern from just the training information?. To answer this, we will consider that similar statistical outcomes across the 3 subsets will produce the same results regardless of the split decision. Differences in the statistical information of the subsets could correlate with the anomalies observed; therefore, the possibility that the model can pick up on this.

F. Outliers detectors.

We will find outliers in our X variables from all subsets by analyzing tables IV and V, the mean, standard deviation, maximums, and minimums. First, we do a general inspection with these operations for all variables. We are looking at a possibly important outlier if the difference between subset stats is considerably different. Having mentioned that, we find our first suspect for further study.

Outlier criteria. The outlier criteria I proposed is that only 1 discrepancy does not raise suspicion; the discrepancies in the train subset are less important than the validation and test discrepancies. It can be a subject of study even if it contains only 1 discrepancy if it has a substantial difference between subsets.

- **Temperature:** the general mean is fairly the same across the subsets; the std is equally high for all of them, but the maximum and minimum trigger the warning. The train contains the coldest instance, with the difference having a minimum of -17.8 degrees Celsius, and the test

TABLE IV
DESCRIPTIVE STATISTICS FOR ALL SUBSETS.

	Subset	test	train	val
Temperature(°C)	mean	13.42	12.88	12.26
	std	9.91	12.10	11.03
	min	-3.10	-17.80	-4.30
	max	30.30	39.40	32.00
Humidity(%)	mean	60.12	58.17	57.07
	std	23.74	20.00	22.95
	min	0.00	0.00	14.00
	max	98.00	98.00	97.00
Wind speed (m/s)	mean	1.60	1.72	1.96
	std	0.91	1.04	1.04
	min	0.00	0.00	0.00
	max	4.30	7.40	5.30
Visibility (10m)	mean	0.61	0.72	0.74
	std	0.29	0.30	0.32
	min	0.04	0.01	0.02
	max	1.00	1.00	1.00
Dew point temperature(°C)	mean	4.98	4.08	2.82
	std	12.79	13.08	12.92
	min	-19.00	-30.60	-22.00
	max	24.50	27.20	23.60
Solar Radiation (MJ/m2)	mean	0.50	0.57	0.61
	std	0.82	0.87	0.91
	min	0.00	0.00	0.00
	max	3.34	3.52	3.52
Rainfall(mm)	mean	0.40	0.14	0.10
	std	2.23	1.04	0.60
	min	0.00	0.00	0.00
	max	29.50	35.00	9.10
Snowfall (cm)	mean	0.01	0.06	0.43
	std	0.04	0.35	1.28
	min	0.00	0.00	0.00
	max	0.30	5.10	8.80

and validation only reach a minimum of -3.1 and -4.3 degrees Celsius, respectively. The train also contains the hottest temperature recorded, with a difference of 39.4 degrees Celsius, while test and val rise to 30.3 and 32, respectively. In theory it is ideal that the train subsets gets the greatest outliers so it could possibly learn from this and then test and validation can be under these confinements but reality does not always fit into therefore the discrepancy makes it a subject of study.

- Humidity gives fairly close outcomes for mean, std, and maximum, with only one minimum discrepancy in the validation subset, never getting a completely dry instance (14% being the driest instance) in comparison to the other that do get a 0. This I consider to be a small discrepancy; therefore, it is not subject to further study.
- Wind speed produces similar outcomes across all stats in exception for train having the windiest instance being 7.4 meters per second in comparison to 4.3 and 5.3, its considerable difference taking into account that the values only ranges from 0 to 7.4 however the data that tell us something to study is the Validation as the mean its greater that the train by itself, this could indicate more windy instances, we will double check.
- Visibility produces similar results across all stats, not a subject of study.
- Dew point temperature almost provides similar results

with an exception in the minimum temperature instances that occurred during train being -30.6 in comparison to test and validation being -19 and -22. Also the mean for validation falls under with 2.8 celsius degrees not that far from test and train being 4.98 and 4.08. It is not consider to be study further.

- Solar radiation outcomes give very similar results in all specs, not a subject to study.
- Rainfall at first look seemed reasonable, with a considerable discrepancy. The maximum rainfall instance happens in train with a 35 mm rainfall anomaly, and in addition, the reason it is going to be further studied is that the test mean gets 0.40mm; as little as it seems, it triples the mean on train and quadruples val mean.
- In snowfall its important to pay attention in if is there a risk on train not getting snow instances. The mean mean for train is 0.06 cm which is low in comparison to validation that has an average 0.43 cm and it turns meaningful comparing their respective std being 0.35 for train and a larger 1.28 std for validation. We can also look at maximum values that continents a considerable difference being 0.3, 5.1 and 8.8 (test, train and validation) cm of snow it will be a subject of study.

We found to be Temperature(°C), Rainfall(mm), Snowfall (cm), Wind speed (m/s) the most dangerous data split since train and validate contains key information of the feature. Therefore we will scope further into by grouping them and comparing by seasons. If the values is not a symptom of seasonality then we can just classify them as expected outliers normal in weather forecast and difficult to predict and wont be looked further into. If it is a cause by seasonality and it got hit right into the split then it should be concluded that there can be another rearrange of data split for future considerations.

TABLE V: Critical statistical values for all subsets

Variable	Seasons	Subset	mean	std	min	max
Temperature($^{\circ}\text{C}$)	1	test	3.25	3.17	-3.10	10.30
		train	-3.09	5.38	-17.80	8.70
		val	1.28	3.44	-4.30	8.30
	2	test	21.03	3.44	14.30	28.70
		train	12.27	6.37	-6.60	29.10
		val	19.16	4.58	11.30	29.40
	3	test	24.45	2.05	20.30	30.30
		train	26.75	4.80	16.30	39.40
		val	25.69	2.99	20.40	32.00
	4	test	4.95	3.42	-1.50	13.20
		train	15.23	6.46	0.30	30.50
		val	2.90	2.83	-3.00	9.50
Rainfall(mm)	1	test	0.19	1.25	0.00	9.50
		train	0.02	0.31	0.00	9.50
		val	0.04	0.38	0.00	3.70
	2	test	0.01	0.09	0.00	1.00
		train	0.20	1.26	0.00	35.00
		val	0.10	0.56	0.00	3.50
	3	test	1.40	4.13	0.00	29.50
		train	0.19	1.30	0.00	24.00
		val	0.13	0.34	0.00	1.50
	4	test	0.00	0.00	0.00	0.00
		train	0.13	0.98	0.00	18.00
		val	0.11	0.94	0.00	9.10
Snowfall (cm)	1	test	0.00	0.00	0.00	0.00
		train	0.25	0.68	0.00	5.10
		val	0.47	1.23	0.00	4.10
	2	test	0.00	0.00	0.00	0.00
		train	0.00	0.00	0.00	0.00
		val	0.00	0.00	0.00	0.00
	3	test	0.00	0.00	0.00	0.00
		train	0.00	0.00	0.00	0.00
		val	0.00	0.00	0.00	0.00
	4	test	0.02	0.08	0.00	0.30
		train	0.00	0.00	0.00	0.00
		val	1.25	2.01	0.00	8.80
Wind speed (m/s)	1	test	1.70	0.87	0.10	3.80
		train	1.92	1.17	0.00	6.70
		val	2.19	1.20	0.00	5.00
	2	test	1.81	0.89	0.30	4.20
		train	1.87	1.08	0.10	7.40
		val	2.03	1.15	0.40	5.30
	3	test	1.54	1.08	0.00	4.30
		train	1.60	0.90	0.00	6.90
		val	1.99	0.87	0.10	4.50
	4	test	1.34	0.71	0.20	3.30
		train	1.50	0.94	0.00	5.70
		val	1.63	0.81	0.40	3.40

TABLE VI
OUTLIERS FOR RAINFALL AND SNOWFALL ACROSS ALL SUBSETS AND SEASONS.

Seasons	Subset	Rainfall(mm)		Snowfall (cm)	
		max	% con evento	max	% con evento
1	test	9.50	2.50	0.00	0.00
	train	9.50	1.90	5.10	19.40
	val	3.70	2.10	4.10	14.60
2	test	1.00	0.80	0.00	0.00
	train	35.00	9.30	0.00	0.00
	val	3.50	4.20	0.00	0.00
3	test	29.50	34.20	0.00	0.00
	train	24.00	6.80	0.00	0.00
	val	1.50	21.90	0.00	0.00
4	test	0.00	0.00	0.30	8.30
	train	18.00	4.90	0.00	0.00
	val	9.10	2.10	8.80	42.70

Scoping even further, we can then get the extreme values for rainfall and snowfall in all sub-sets in all seasons. Reading our VI table, we have spotted some important outliers. Reading the table for winter, the most important thing is that we can see the maximum value falls under train, which is good; however, we can spot that the frequency of each instance in train is lower. For spring, it does gets the rainiest instances, with a frequency of 9.3 that surpasses test by a lot. Test and validation this season could throw some models, as it could be expecting the rainiest instances around spring and then validating and testing under- or overestimating. For summer, it worries the other way around: validation and test get more than 20% of the instances of rain. Lastly, for autumn, the train does get the maximum value and highest frequency, making it the most representative subset. For snowfall, we only check winter and autumn, as it only snows during these seasons; for winter, it is reasonable as the train gets the maximum value and the highest frequency; however, in autumn it provides an interesting signal. In autumn, the train does not get any snowfall; test and validation do, and even more extreme, validation has a maximum of 8.8 cm, the highest recording in all data set, with more than 42% of instances snowing. The general conclusion is that even with a seasonality split, the train does not always get the most representative content, and the question then is: is it really enough provided to the model to pick up on this data?.

Finally, looking at our Table VII we have created the same analysis system, reading by season by subset the statistics. The biggest difference and most worrisome is the train subset for Spring, the mean of rental bikes falls behind by a lot compared with test and val for the same season; the std also falls behind. Then the greatest notice is that in Summer the training subset does get the higher mean; it could mean that it represents the higher outliers.

G. EDA Closure.

We will now try to give a conclusion based on all the results above. First, we have spotted our most problematic features that, due to the split, may present trouble; we can visualize our individual data collected better and try to pinpoint our findings

TABLE VII
RENTED BIKE COUNT DATA FOR ALL SEASONS AND SUBSETS.

Seasons	Subset	Statistic measurements			
		mean	std	min	max
1	test	289.88	209.44	7	929
	train	217.81	142.35	3	937
	val	301.61	179.44	32	784
2	test	1220.16	740.28	134	3251
	train	702.50	587.18	2	3130
	val	1183.24	703.37	25	3123
3	test	679.06	646.26	10	2836
	train	1068.03	691.68	9	3556
	val	773.21	516.11	30	2080
4	test	689.07	373.12	95	1708
	train	961.09	627.05	2	3298
	val	526.03	348.47	24	1671

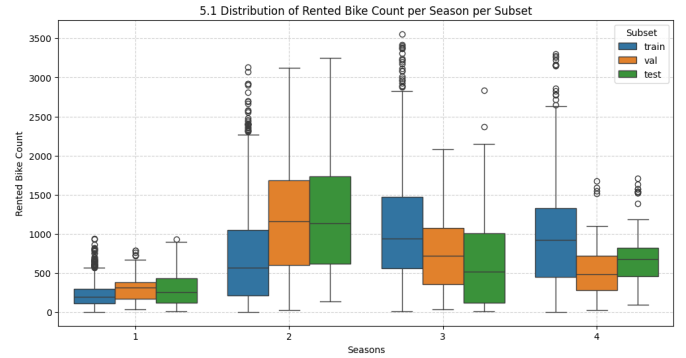


Fig. 8. Boxplot for rental bikes per season per subset.

with graphics 5. 5.1 Repeating the theoretically proposed sweet point is to have the biggest number of outliers in the train section so it demonstrates that it is representative; what we essentially found for the rented bike count was that it was almost true. The problems can be looked in the spring when we notice that even if the train gets the most outliers val and test long whiskers reach these train outliers and for Autumn even if the test and validation falls under the train boxplot this is also not so good we would prefer similar boxes and whiskers lengths as in winters with the fewest outliers possible as in summer.

5.2 , 5.3 , 5.4 , 5.5 For temperature, the overall look is ideal; the data didn't spike before but had to make sure visually as the snow outliers gave me the impression that it could also be affected to temperature, if it wasn't spotted on our tables and visually we conclude the train section do get the representative data and there is not an issue. The rented bike count confirms our reading by season. For wind speed, we find a visually better outcome than expected as the box plots its self and whiskers looks similar and train does get the most outliers, finally the snow and rainfall instances do rise some heavy concern as the bar graph represents the frequency of instances the subset has and for validation it contains a very high frequency of snow, for rainfall it is most acceptable but we also had pinpointed the standard deviation of test rainfall

data was the largest which potentially could indicate a more distribution difficult to predict.

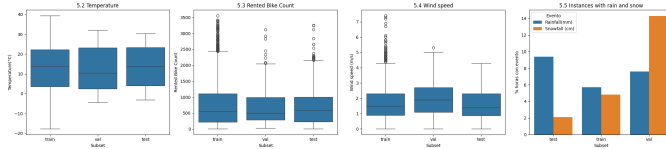


Fig. 9. Boxplot for critical outliers.

We can finally produce a correlation analysis that is represented through this 6.1 heatmap. We are looking for the highest correlation features to try and determine if any of these variables can be dropped due to a high correlation between each. Right off the bat, we can spot the dew temperature and temperature itself with a 0.92 correlation; they move similarly across the whole dataset, and at first it was a logical decision to drop them as they almost target and track similar things. However, I have decided not to drop this entire feature one or the other due to their independent correlation with others. If you notice, temperature has a low correlation with Humidity of 0.15, while dew point temperature has a mild correlation of 0.52. The same thing was spotted with solar radiation: dew point has a lower correlation of 0.10 and temperature a greater correlation of 0.36. The different correlations each variable possesses with another could be an advantage for certain models or procedures, so we are going to leave it as is. Every other feature does not possess an interesting correlation.

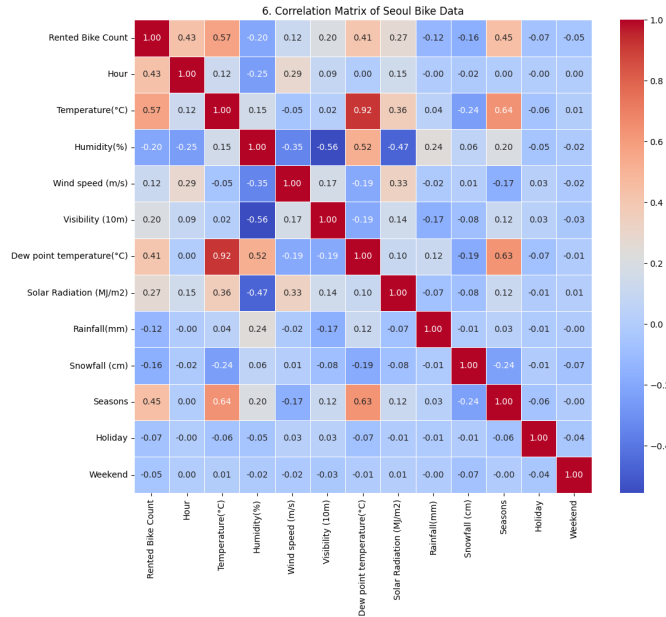


Fig. 10. HeatMap.

The final verdict is that the dataset will be left as is, since imperfection it will always expected, especially when forecasting this kind of problem; the specific selection of data will create a biased output which we will not be able to take advantage of, as we have managed to perfect our problem.

What is important to document is that all of this imperfection, although acceptable to proceed, must be explained rather than avoided so it can provide possible explanations further in this project. Valuing my strategy for split, the seasonality-based division did succeed in giving most of the subsets representation for all seasons. It is important to mention that it failed during the autumn season since the training subset won't get any of the most extreme values, therefore creating the actual risk of training a model that has a bias due to omission. So no further study will be made; no variables will be removed or created and transformed. We are ready to enter the discussion of the model design.

IV. ML MODELS

A. The choosing of the model

The EDA established two facts that give us the clue for a preferred model choice. One is the relationship between several features and the non-linear demand. It contains many behaviours that seem cyclic or approach fuzzy classes, and the dependent variable itself does not seem to hold a trend to be described linearly. The other fact is that the ETL output we provided is not friendly with a linear model: take the hours as integers from 0-23; that is not thought of as cyclic, and 23 won't be mathematically thought of as near 0; same thing for seasons. The expected output of a linear regression will be a not-so-well-fitted line, as all of these factors prove to be of theoretical complexity for the regression model. We will produce, however, a linear regression model by hand. The idea will be that our starting point will be the final metrics and scores for the linear regression trained with batch gradient descent without a learning framework for academic purposes to learn more about the construction of the model. This first model is chosen because it is the easiest hypothesis to combine all 12 features we are creating; it will be the most natural first attempt. Secondly, we can run an OLS regression model to back up the performance of the model on the expected outputs and evaluate how much the linear model could really explain. This will also provide the expected records to beat; I have indicated that other models might fit best to this problem, and then it shall be compared to.

Then our next models will compare something different to the linear thought. The proposals are a decision tree that explores regressor fit instead of classification; this will be fitted first because it makes the change in assumption easy to follow instead of fitting one constant fixed weight per feature. In theory, I think this model will be able to pick up and isolate specific peaks and patterns that a linear coefficient could not represent. Then, by the same logic, I will try once again through a random forest regression with many more trees to reduce the variance that causes a single deep tree to overfit and punish this. We are left then with 4 final models: the linear model by hand that will provide the floor to compare to, the OLS regression that will back up the performance of the linear and then the decision tree and random forest that will try to prove the question left open by the exploratory analysis. Since every model receives the same training subset, carrying

the same documented gaps in autumn snowfall and summer rainfall, any difference in their results cannot be attributed to the split. The comparison therefore measures the cost of the linear assumption specifically.

B. Linear Regression Results

The initial setting of the model was set to train for 300 epochs with a learning rate of 0.05, using batch gradient descent over the 7,896 training instances. Parameters(Bias and weight) at first were initialised at random in the range minus 0.01 to 0.01 with a fixed seed, this decision was made to try and avoid to hit minimum locals. Graph 7.1 shows the cost on both the training and validation subsets at every epoch. Its visible thatt both of the curves descend together and remain close throughout, with no widening gap between them. I consider this to be a case of non overfitting nor underfitting, a model that had begun memorising the training subset would show the training cost continuing to fall while the validation would diverge from it. The absence of that pattern is expected for a linear model, which has too little capacity to memorise 7,896 instances through twelve weights. There was no need to implement a strategy that could punish overfitting. The final metrics are the model reaches an Rsquared of 0.3523, an RMSE of 505.30 bikes and an MAE of 361.25 bikes. the training mean reaches an RMSE of 628.21 and an Rsquared of minus 0.001. Meaning the model explains roughly 35% of the variance in hourly demand. It is, however, far from an accurate prediction since the average error of 505 bikes is substantial against a training mean of 701 bikes per hour. From the graph 7.2 we can see the model has a harsh time fitting the linear model. This is the behaviour expected backed up by our eda, the non linear relationship has a maximum that can be forced through a straight.

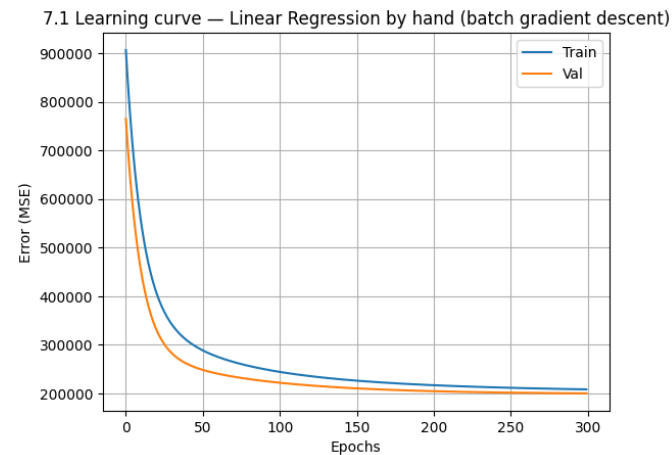


Fig. 11. HeatMap.

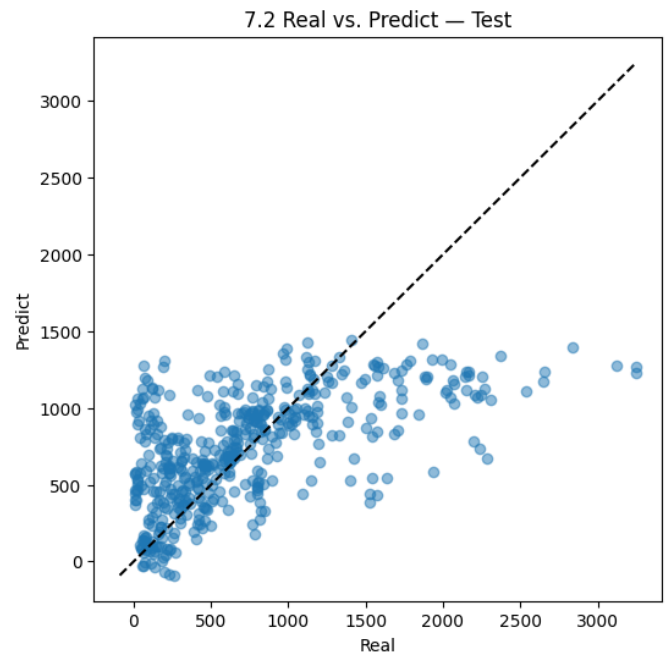


Fig. 12. HeatMap.

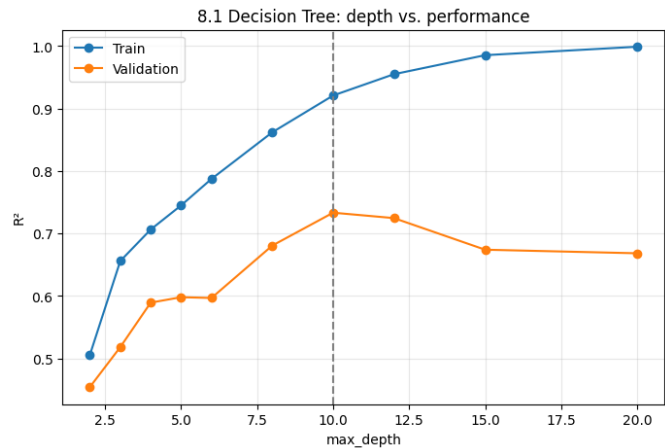


Fig. 13. HeatMap.

V. PRINCIPAL COMPONENT ANALYSIS (PCA)

VI. MODEL CONSTRUCTION (MANUAL)

VII. MODEL CONSTRUCTION (FRAMEWORK)

VIII. RESULTS AND ANALYSIS

IX. VERSIONS AND COMPARISON

X. CONCLUSION

REFERENCES

- [1]
- [2]